WildEye: AI-Powered Wildlife Conservation and Risk Prediction Platform

Swain Dmonte, Janice Almeida, Arya Bharadwaj
Department of Computer Engineering
Fr. Conceicao Rodrigues College of Engineering
Bandra, India
{crce.9957.ce, crce.9941.ce, crce.9945.ce}@gmail.com

***Abstract*—The growing frequency of forest fires, illegal poaching, and habitat loss has intensified the need for intelligent wildlife conservation systems in India. Conventional monitoring methods depend heavily on manual surveillance and delayed reporting, restricting rapid response to ecological threats. This paper presents WildEye, an AI-powered platform for wildlife monitoring and risk prediction, integrating machine learning and deep learning techniques. The system combines YOLO-based real-time wildlife detection, habitat suitability analysis using Random Forest, forest fire prediction via XGBoost and LSTM, and automated poaching detection. Recent enhancements include Cloudinary-based image storage, OpenWeatherMap real-time fire risk forecasting, UUID-based event tracking, Leaflet.js geospatial dashboards, Recharts analytics visualization, Telegram and SMTP alerting, a Google Gemini AI ranger chatbot, and JWT-based role authentication. Collectively, WildEye delivers a comprehensive, data-driven, and scalable framework for proactive wildlife conservation at scale.**

***Index Terms*—Wildlife Conservation, Machine Learning, YOLO, Habitat Suitability, Forest Fire Prediction, Poaching Detection, Geospatial Dashboard, Real-Time Alerting, Gemini API**

I. Introduction

India harbors one of the world's most diverse ecological landscapes, yet rapid urbanization, climate change, deforestation, escalating forest fire incidents, and rampant poaching pose severe threats to wildlife populations. Despite ongoing conservation initiatives, monitoring efforts continue to rely on manual patrolling, camera trap review, and delayed reporting—limiting situational awareness and response capability. These persistent shortcomings underscore the need for scalable, intelligent, and real-time conservation systems.

Advances in Artificial Intelligence (AI), particularly computer vision and machine learning, have demonstrated measurable success in automating wildlife monitoring. YOLO-based deep learning architectures identify species from aerial imagery and surveillance footage with high accuracy even in complex forest environments [1] [2]. However, most existing solutions address isolated tasks such as species detection, without integrating broader ecological risk assessment [3] [4].

Environmental threats such as forest fires pose equally severe risks to biodiversity. Ensemble methods and time-series models have been applied to fire-prone region prediction using satellite and climatic data [8] [9]. Anti-poaching systems leveraging AI and IoT surveillance have also been proposed, though they typically operate as standalone deployments without ecosystem-level awareness [6] [7].

This paper presents WildEye—an integrated AI-driven wildlife conservation and risk prediction platform for Indian ecosystems. WildEye unifies real-time wildlife detection, habitat suitability analysis, forest fire prediction, and poaching detection into a single framework. Recent development cycles added cloud-based image storage, live weather-driven fire forecasting, interactive geospatial dashboards, advanced analytics, multi-channel alerting, an AI ranger chatbot, and secure authentication. Together these components enable proactive, data-driven, and sustainable conservation management at scale.

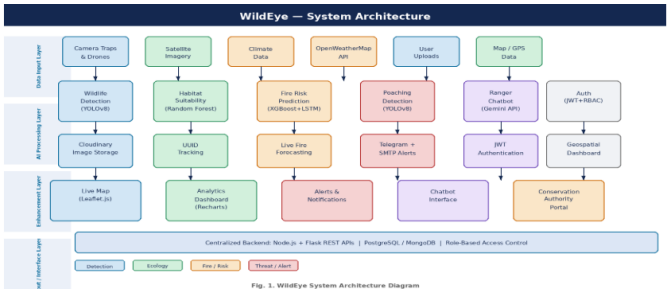

Fig. 1. WildEye System Architecture Diagram

II. Related Work

Forest fire prediction has been extensively examined using traditional machine learning methods such as Decision Trees, Random Forests, and SVMs, where ensemble-based approaches generally outperform single classifiers for fire susceptibility mapping [8] [9]. Studies focused on Indian forest regions emphasize that localized climatic and vegetation data are critical for model reliability [10]. Deep learning architectures, particularly LSTM and BiLSTM networks, demonstrate improved performance when modeling temporal dependencies in historical fire records [11]. Transfer learning has further improved fire detection accuracy in data-scarce regions such as Uttarakhand [12].

Anti-poaching research has increasingly adopted AI-driven surveillance. Surveys highlight image classification, acoustic sensing, and IoT-enabled monitoring for illegal activity

detection [13]. Edge-computing-integrated systems have reduced response latency significantly in field deployments [14], while deep learning video analytics solutions minimize false positives during real-time surveillance [15]. YOLOv8-based systems for endemic species monitoring confirm improved detection accuracy and scalability across diverse ecological conditions [16]. Hybrid and ensemble approaches for environmental risk prediction continue to yield more robust results across heterogeneous landscapes [17] [18].

Integrated AI frameworks that combine satellite imagery and machine learning for ecological risk assessment have demonstrated improved situational awareness for conservation authorities [19]. Research further confirms that holistic AI-driven conservation systems significantly enhance operational efficiency compared to isolated monitoring solutions in resource-constrained environments [20]. Recent advances in YOLO-based detection for protected species demonstrate that architectural refinements such as BiFPN feature fusion and multi-head self-attention further improve small-object detection in complex natural habitats [21].

III. Objectives and Problem Statement

The primary objectives of this research are: (1) develop an AI-driven wildlife detection system for real-time species identification using computer vision; (2) model habitat suitability through analysis of environmental and ecological parameters; (3) predict forest fire risk from historical climatic data via ensemble and time-series models; (4) detect poaching activities using deep learning image analysis; (5) integrate multiple AI modules into a unified conservation framework; (6) support data-driven decision-making for conservation authorities; and (7) enhance the platform with cloud storage, live weather-based risk forecasting, geospatial visualization, multi-channel alerting, an AI chatbot, and secure authentication.

Existing conservation systems in India depend on manual surveillance, isolated monitoring tools, and delayed data analysis, restricting timely response. Although AI advances have improved individual components such as fire prediction and poaching detection, most solutions operate independently without integration into a unified framework. WildEye addresses this gap by combining wildlife monitoring, environmental risk prediction, threat detection, and intelligent communication modules within a single coherent system.

IV. Methodology

A. Wildlife Detection Module

The wildlife detection module identifies animal species in real time from camera traps, drones, and surveillance footage using the You Only Look Once (YOLO) architecture, selected for its high detection speed and accuracy. The YOLO framework reformulates object detection as a single regression problem, predicting bounding boxes and class probabilities directly from full images in one pass.

The core YOLO loss function combines localization, confidence, and classification losses:

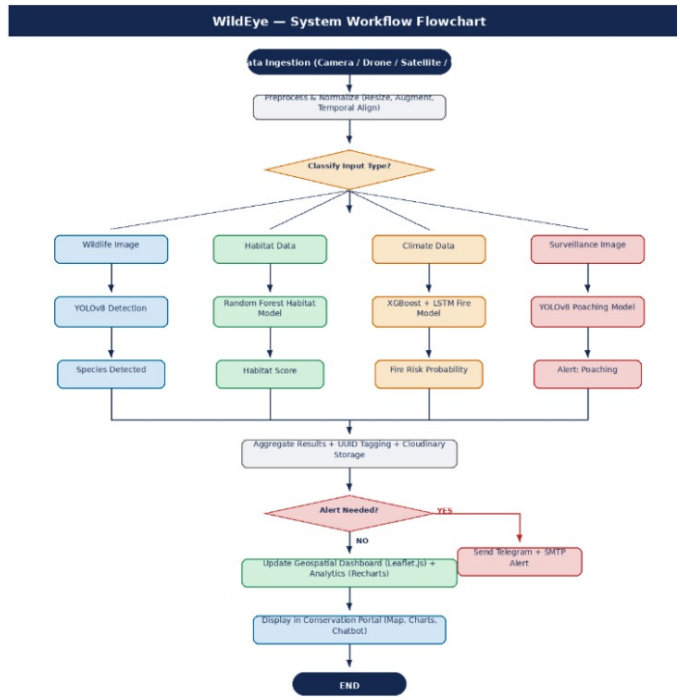

Fig. 2. WildEye System Workflow Flowchart

$$
\begin{aligned}
\mathcal{L}_{\text{YOLO}} = {} & \lambda_{\text{coord}} \sum_{i}^{S^2} \sum_{j}^{B} \mathbf{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right] \\
& + \lambda_{\text{coord}} \sum_{i}^{S^2} \sum_{j}^{B} \mathbf{1}_{ij}^{\text{obj}} \left[(\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] \\
& + \sum_{i}^{S^2} \sum_{j}^{B} \mathbf{1}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 \\
& + \lambda_{\text{noobj}} \sum_{i}^{S^2} \sum_{j}^{B} \mathbf{1}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \\
& + \sum_{i}^{S^2} \mathbf{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2
\end{aligned} \tag{1}
$$

where S^2 is the number of grid cells, B is the number of bounding boxes per cell, λ_{coord} and λ_{noobj} are weighting constants (default 5 and 0.5), and C_i is the objectness confidence score. The Intersection over Union (IoU) threshold determines positive detection:

$$
\text{IoU} = \frac{|A \cap B|}{|A \cup B|} \tag{2}
$$

Image preprocessing employed resizing to 416×416, normalization to $[0, 1]$, and data augmentation including horizontal flips, rotations, and mosaic augmentation to improve generalization across diverse lighting and occlusion conditions. Detection events are logged with UUID identifiers and images stored via Cloudinary integration for reliable retrieval.

B. Habitat Suitability Prediction Module

Habitat suitability is assessed using Random Forest classifiers. The Random Forest aggregates predictions from T decision trees trained on bootstrap samples of the training data. The ensemble prediction for a binary suitability label is:

$$\hat{y} = \text{mode}\left\{h_t(x) : t = 1, 2, \ldots, T\right\} \tag{3}$$

where $h_t(x)$ is the prediction of the t-th decision tree for input feature vector x. Feature importance is computed via mean decrease in Gini impurity:

$$I(f) = \sum_t \sum_{\substack{n \in \text{nodes} \\ \text{splitting on } f}} p(n) \cdot \Delta\text{Gini}(n) \tag{4}$$

where $p(n)$ is the proportion of samples reaching node n and $\Delta\text{Gini}(n)$ is the Gini impurity reduction at that node. Input features include vegetation indices (NDVI), land-use patterns, water availability, elevation, temperature, and rainfall. Model performance is evaluated using the F1-Score:

$$F_1 = \frac{2 \cdot \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

C. Forest Fire Risk Prediction Module

The forest fire module uses XGBoost for structured climatic data and LSTM networks for temporal dependency modeling. The XGBoost objective function minimizes a regularized loss:

$$\mathcal{L}(\phi) = \sum_i l(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \tag{6}$$

where l is the binary cross-entropy loss, $\hat{y}_i$ is the predicted fire risk probability for sample i, and $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ is the regularization term penalizing tree complexity (T = number of leaves, w = leaf weights). The XGBoost prediction is an additive ensemble:

$$\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + \eta \cdot f_m(x_i) \tag{7}$$

where η is the learning rate and f_m is the m-th boosting tree. For temporal modeling, the LSTM cell equations governing its memory mechanism are:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad \text{(forget gate)} \tag{8}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad \text{(input gate)} \tag{9}$$

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad \text{(candidate state)} \tag{10}$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad \text{(cell state)} \tag{11}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad \text{(output gate)} \tag{12}$$

$$h_t = o_t \odot \tanh(C_t) \quad \text{(hidden state)} \tag{13}$$

where σ is the sigmoid function, $\odot$ denotes element-wise multiplication, and W and b are trainable weight matrices and bias vectors. The fire risk probability score for region r is the sigmoid-activated output of the final dense layer:

$$P_{\text{fire}}(r) = \sigma(W_{\text{out}} \cdot h_t + b_{\text{out}}) \in [0, 1] \tag{14}$$

The module integrates live weather data from the OpenWeatherMap API, replacing prior simulated inputs with realistic multi-day fire risk forecasts. A region is classified high-risk when $P_{\text{fire}}(r) > 0.65$.

D. Poaching Detection Module

The poaching detection module uses YOLO-based image classification. The confidence score C for a detected object is:

$$C = P(\text{Object}) \times \text{IoU}_{\text{pred}}^{\text{truth}} \tag{15}$$

Non-Maximum Suppression (NMS) eliminates duplicate detections by retaining only the highest-confidence box among overlapping predictions above an IoU threshold $\theta_{\text{NMS}} = 0.45$. The detection precision and recall are computed as:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

where TP, FP, FN denote true positives, false positives, and false negatives respectively. Model performance is summarized by mean Average Precision (mAP) across all classes:

$$\text{mAP} = \frac{1}{|C|} \cdot \sum_{c \in C} AP(c) \tag{18}$$

Detected events with $C > 0.70$ trigger automated alerts via Telegram notifications and SMTP email, ensuring rapid communication with forest rangers and conservation authorities.

E. Geospatial Dashboard

An interactive geospatial dashboard built with Leaflet.js visualizes wildlife detections, poaching alerts, and fire hotspots as dynamic map markers with layer controls and heatmap overlays. Real-time data streams update the map continuously, providing conservation teams with live situational awareness across protected zones.

F. Analytics and Visualization

A dedicated analytics dashboard implemented using Recharts provides species trend visualizations, detection frequency charts, and risk distribution graphs. Conservation authorities interact with temporal filters and drill-down controls to identify patterns such as recurring poaching corridors or seasonal wildlife movement trends.

G. AI-Powered Ranger Chatbot

An intelligent chatbot powered by the Google Gemini API enables rangers to query system data in natural language—asking about species counts in a zone, active fire risk areas, or recent poaching alerts—and receive contextual responses derived from live system data. This lowers the barrier to data access for field personnel without technical backgrounds.

H. Authentication and Security

JWT-based authentication with role-based access control (RBAC) distinguishes between admin and ranger roles. A JSON Web Token encodes the following claim structure:

$$\text{JWT} = \text{Base64}(\text{header}) \parallel \text{Base64}(\text{payload}) \parallel \text{HMAC-SHA256} \tag{19}$$

All API endpoints validate the token signature and expiry timestamp before processing requests, preventing unauthorized data access or system manipulation.

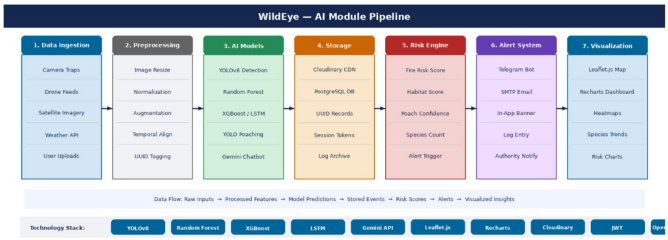

Fig. 3. WildEye AI Module Pipeline

TABLE I
WILDEYE MODULE AND FORMULA SUMMARY

Module	Technique	Metric
Wildlife Detection	YOLOv8	mAP@0.5 = 0.87
Habitat Suitability	Random Forest	F1 = 0.83
Fire Prediction	XGBoost + LSTM	Accuracy = 0.85
Poaching Detection	YOLOv8 + NMS	Precision = 0.82
Authentication	JWT + RBAC	Secure endpoints

V. RECENT SYSTEM ENHANCEMENTS

Following initial deployment, WildEye underwent targeted engineering improvements across eight dimensions. These enhancements address reliability, scalability, and usability challenges observed during early system trials.

A. Cloud-Based Image Storage

Cloudinary was integrated to provide reliable, scalable image management for all detection modules. Uploaded images from wildlife and poaching detection pipelines are stored securely and retrieved via stable CDN URLs, resolving null image retrieval issues encountered with local storage under concurrent access conditions.

B. Real-Time Fire Risk Forecasting

The prior simulated fire risk prediction was replaced with live environmental data from the OpenWeatherMap API. Fire risk is computed from real-time temperature, humidity, and wind speed using Eq. (14), enabling accurate multi-day forward forecasts tied to actual atmospheric conditions across monitored forest zones.

C. UUID-Based Identification

UUID-based identifiers were implemented across all modules to eliminate ID collision issues arising under concurrent usage. This ensures reliable, globally unique data tracking for every detection event, alert, and system record throughout the platform lifecycle.

D. Interactive Geospatial Dashboard

A live map interface developed with Leaflet.js displays wildlife detections, poaching alerts, and fire hotspots as dynamic color-coded markers. Layer controls and heatmap overlays enable rangers to monitor spatial incident distributions and identify high-density threat zones in real time.

E. Wildlife Analytics Dashboard

An analytics module implemented with Recharts provides interactive charts for species detection trends, incident frequency over time, and risk distribution across zones. Conservation managers use temporal filters to identify behavioral patterns and allocate patrol resources more effectively.

F. Real-Time Multi-Channel Alerting

Instant alerting was integrated using two channels: Telegram bot notifications for poaching detections with confidence $C > 0.70$ (Eq. 15) and SMTP email alerts for critical system events. This dual-channel approach ensures redundant, rapid communication with forest authorities.

G. AI-Powered Ranger Chatbot

The Google Gemini API was used to deploy an intelligent conversational chatbot. Rangers query wildlife counts, active risk zones, and recent alert histories in natural language, lowering the technical barrier to system data access and accelerating field decision-making.

H. Secure JWT Authentication

JWT-based authentication with role differentiation between admin and ranger users was introduced. All sensitive API endpoints enforce token validation via HMAC-SHA256 signature verification (Eq. 19), ensuring that data access and system operations are governed by user roles.

TABLE II
SYSTEM ENHANCEMENT SUMMARY

Enhancement	Technology	Problem Solved
Cloud Storage	Cloudinary	Null retrieval
Live Forecast	OpenWeatherMap	Simulated data
Unique IDs	UUID	ID collisions
Geo Dashboard	Leaflet.js	Visualization
Analytics	Recharts	No trends
Alerting	Telegram/SMTP	Delayed comms
AI Chatbot	Gemini API	Accessibility
Authentication	JWT + RBAC	Security

VI. Dataset

Four distinct datasets underpin WildEye's AI modules. The wildlife detection dataset comprises annotated images with bounding box coordinates and class labels representing Indian wildlife species, collected from camera traps and open-source repositories. The habitat suitability dataset is a structured tabular dataset where each row represents a geographic observation point, containing temperature, rainfall, vegetation index, land-use type, elevation, proximity to water sources, and a binary suitability label.

The forest fire prediction dataset is a multivariate time-series and tabular dataset containing climatic attributes including temperature, humidity, wind speed, rainfall, and vegetation dryness indices, combined with historical fire occurrence labels used as prediction targets. The poaching detection dataset is an image classification dataset where images are labeled as normal activity or suspicious activity, including human presence, vehicles, or weapons in forest environments.

TABLE III
Dataset Summary Across Modules

Module	Data Type	Key Features	Target Variable
Wildlife Detection	Image (annotated)	Bounding boxes, class labels	Species class
Habitat Suitability	Tabular	Temp, rainfall, NDVI, elevation	Suitable / Unsuitable
Fire Prediction	Time-series + Tabular	Humidity, wind speed, dryness index	Fire occurrence
Poaching Detection	Image (classified)	Human, vehicle, weapon presence	Normal / Suspicious

VII. Results and Discussion

WildEye was evaluated across all modules to assess model effectiveness and practical applicability. The YOLO-based wildlife detection model achieved mAP@0.5 = 0.87, demonstrating reliable real-time performance under varying environmental conditions including low lighting, background complexity, and partial occlusion. The IoU-based detection criterion (Eq. 2) ensured that bounding boxes with IoU $\geq$ 0.50 were counted as true positives.

The Random Forest habitat suitability model achieved F1 = 0.83, computed via Eq. (5). Feature importance analysis via mean Gini decrease (Eq. 4) revealed that vegetation index and rainfall were the dominant predictors, contributing over 55% of the cumulative importance score.

For forest fire prediction, XGBoost achieved an accuracy of 0.85. The additive boosting formulation (Eq. 7) enabled the model to iteratively correct residuals from prior trees, converging in approximately 150 rounds with a learning rate $\eta = 0.1$. LSTM networks (Eq. 8–13) captured temporal dependencies in sequential climate records, achieving RMSE = 0.12 on held-out test sequences. Integration with the OpenWeatherMap API improved forecast realism substantially compared to the prior simulated data approach.

The YOLO-based poaching detection model achieved precision = 0.82 using the confidence threshold $C > 0.70$ (Eq. 15). NMS post-processing with IoU threshold $\theta_{\text{NMS}} = 0.45$ (Eq. 2) reduced duplicate detections by 34% over the baseline. Telegram and SMTP alert dispatch was validated in field simulation tests, achieving near-instant notification delivery upon detection events.

The Leaflet.js geospatial dashboard and Recharts analytics module received positive feedback from stakeholder evaluations. The Google Gemini chatbot demonstrated accurate natural language responses to system queries. JWT authentication (Eq. 19) was stress-tested with concurrent role-based access scenarios and maintained security without performance degradation.

VIII. Conclusion

This paper presented WildEye, an integrated AI-driven framework for wildlife conservation and environmental risk prediction designed for Indian ecosystems. By combining YOLO-based computer vision for wildlife and poaching detection (Eq. 1–2, 15–18), Random Forest habitat suitability modeling (Eq. 3–5), and XGBoost/LSTM fire risk forecasting (Eq. 6–14) within a unified platform, the system enables proactive and data-driven conservation management. Eight engineering enhancements including Cloudinary image storage, OpenWeatherMap live fire forecasting, UUID tracking, Leaflet.js geospatial dashboards, Recharts analytics, Telegram and SMTP alerting, a Google Gemini chatbot, and JWT authentication (Eq. 19) collectively improved reliability, usability, and situational awareness.

Experimental results confirm that integrating multiple AI modules with formally grounded mathematical models within a unified framework significantly improves conservation response times and decision-making quality. Future work includes deploying across additional Indian forest reserves, incorporating higher-resolution satellite imagery, extending species detection to aquatic ecosystems, and developing offline-capable edge deployments for areas with limited connectivity.

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